



United International University (UIU)
Dept. of Computer Science & Engineering (CSE)
Mid-Term Exam Trimester: Spring 2025

Course Code: CSE 4325 Course Title: Microprocessors and Microcontrollers
 Total Marks: 30 Duration: 1 hour 30 minute(s)

Any examinee found adopting unfair means would be expelled from the trimester/ program as per UIU disciplinary rules.

Question 1: Answer all the questions. (7.5 Marks)

Transfer of bus control from processor to device takes **0.2 μ s(microseconds)**. Transfer of bus control from device to processor takes **0.4 μ s(microseconds)**.

a.	If one of the input/output devices employs DMA in Cycle Stealing Mode and takes 62,960,800 ns to transfer some data at a transfer rate of 96 KB/s with 8 Bytes of data being transferred at once, find out the total size of the data to be transferred in Bytes .	[4]
b.	Suppose, you will be transferring the total data (<i>found from (a)</i>) now in both Burst Mode and Cycle Stealing Mode. For the first one-third of the bytes, you use Burst Mode and for the rest , you use Cycle Stealing Mode . Assuming that cycle stealing mode transfers 8 Bytes at a time and both modes use 96 KB/s transfer rate, how long will it take to transfer the whole data?	[3.5]

Question 2: Answer all the questions. (7.5 Marks)

a.	Execution of a signed addition instruction (76DAH + 3867H) occurred. What would be the values of the sign flag (SF), parity flag (PF), auxiliary flag (AF), carry flag (CF), and overflow flag (OF)? Provide a brief explanation for your answer.	[5.5]
b.	State two differences between the Primary memory and Secondary memory.	[2]

Question 3: Answer all the questions. (7.5 Marks)

a.	CS = 25A1 H, DS = 4132 H, SS = 3123 H, ES = 5132 H IP = 2451 H, SI = 134B H, SP = 2354 H I) Find how many physical address slots exist that are overlapping between Code segment and Stack segment ? II) Can the Data Segment be placed after the Stack Segment without overlapping with the Extra Segment ? Explain with mathematical reasoning.	[2.5+2.5]
----	--	------------------

b.	RAM X has two third times the capacity of RAM Y. The total memory capacity of both RAMs combined is 15 MB. If RAM X has a 22-bit address bus, determine the size of its data bus.	[2.5]
<u>Question 4: Answer all the questions. (7.5 Marks)</u>		
a.	You need to design a system where a microprocessor with a 18-bit address bus and 16-bit data bus is interfaced to a 48 KB RAM system using Linear decoding. Each RAM chip has a 12-bit address bus and a 16-bit data bus. Draw a block diagram showing the microprocessor, RAM chips, and linear decoding logic. Provide the 3rd, 5th and 7th address of the last three RAM chips.	[4+3.5]